

FINAL EXPLANATION: BIOLOGY GRADE 11

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 11: GROWTH AND DEVELOPMENT IN ANIMALS (3.2)

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Use this document as a model for your own Final Explanation. Your task is to answer the driving question: How does an animal outgrow its own hard shell, and why do some animals radically change shape while others just grow bigger? Build a CER (Claim, Evidence, Reasoning) poster with an annotated graphical model. Anchor every section in the phenomenon: the empty, split exuvia beside a larger, newly emerged grasshopper or cicada on a maize leaf or tree bark in your school garden, and the caterpillar that becomes a pupa and then a butterfly. Use evidence from your six lessons, such as growth curves, observations of exuviae, and hormone diagrams. Write in your own words and label all graphs and diagrams clearly.

1. Claim: Answering the Driving Question

State in 2-3 sentences how an animal outgrows a hard shell and why some animals change shape completely while others only grow bigger. Mention the two growth patterns and the role of hormones.

An insect outgrows its hard shell by shedding it in a process called ecdysis (moulting). The shell cannot stretch, so the insect grows in sudden steps each time it sheds the old shell and swells before the new one hardens. Animals with a hard exoskeleton, like grasshoppers and cicadas, grow in steps (intermittent growth). Vertebrates like cattle, goats and humans grow continuously, giving an S-shaped curve. Hormones control the process. Brain hormone (PTTH) triggers ecdysone to cause moulting, and juvenile hormone (JH) decides whether the insect stays a larva or nymph or changes into an adult. Insects with complete metamorphosis, like butterflies, change shape totally because JH drops, while those with incomplete metamorphosis, like grasshoppers, look like small adults all along.

2. Evidence: Growth Patterns and the Exuvia

Describe the evidence from the empty exuvia and from growth curves. Compare the graph of an insect (such as a grasshopper nymph in Nairobi or Kisumu) with the graph of a vertebrate (such as a calf or a child). Explain what the shape

The exuvia we found on the maize leaf was crunchy, perfectly shaped and split down the back. Next to it was a grasshopper clearly larger than the shell. This shows the shell was left behind and the animal grew bigger than it, so the insect did not peel like sunburned skin. It left through the split. When we plotted body size against time for a grasshopper, the graph looked like a staircase: size stayed flat while the hard shell was in place, then jumped up at each moult. This is intermittent growth. When we plotted a vertebrate such as a calf or a child, the graph was a smooth S-shaped (sigmoid) curve with four phases: a lag phase where growth is slow, an exponential phase where growth is rapid, a deceleration phase where growth slows down, and a stationary phase where size stays steady in the adult. A vertebrate has an internal skeleton that grows with the body, so no shell must be shed.

of each graph shows.	
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3. Evidence: Types of Metamorphosis	
Use photos or samples of a butterfly life cycle and a grasshopper life cycle. Describe the stages of complete and incomplete metamorphosis and explain how the caterpillar-to-butterfly change is different from a grasshopper's growth.	A butterfly (for example, the common African monarch seen in Kericho tea farms) goes through complete metamorphosis: egg, larva (caterpillar), pupa, adult. The caterpillar looks nothing like the adult. It eats leaves and moults several times, then becomes a motionless pupa. Inside, the body is reorganised under hormone control into wings and adult organs, so it does not simply melt and rebuild at random. A grasshopper goes through incomplete metamorphosis: egg, nymph, adult. The nymph looks like a small adult without full wings and grows through several moults until it is an adult. There is no pupa stage. So both groups moult, but only the butterfly has a pupa stage and a completely different body shape in the adult.

4. Reasoning: Hormonal Control of Moulting and Metamorphosis	
Explain step by step how the hormones PTTH, ecdysone and juvenile hormone control moulting and metamorphosis. Link this reasoning back to your claim and evidence and to the exuvia and the pupa.	Moulting is controlled by a chain of hormones. First, the brain releases Brain Hormone (PTTH). PTTH stimulates the prothoracic glands to release ecdysone. Ecdysone triggers the insect to moult, which is why the exuvia splits and the animal climbs out larger. Juvenile hormone (JH) decides what kind of moult happens. When JH concentration is high, ecdysone causes a moult that keeps the larval or nymph form, so the caterpillar or nymph just gets bigger. When JH decreases, ecdysone causes pupation and then metamorphosis into the adult. This explains our evidence: the staircase graph shows moults caused by ecdysone, the grasshopper stays similar in shape because JH is present through the nymph stages, and the butterfly changes shape completely because JH falls so the caterpillar becomes a pupa and then an adult. So the changes are controlled by hormones and are not random melting.

5. Annotated Graphical Model and Conclusion	
Describe the annotated model you would draw on your poster. Include both growth curves, the hormone sequence, and labels for the exuvia and the life cycles. Finish with a conclusion that answers the driving question and corrects one misconception.	My poster model has two graphs side by side. On the left is a staircase graph labelled 'Insect: intermittent growth' with arrows at each step marked 'ecdysis: ecdysone peak, old exuvia shed'. On the right is an S-shaped curve labelled 'Vertebrate: continuous growth' with the lag, exponential, deceleration and stationary phases marked. Below the graphs is a flow diagram: Brain, PTTH, prothoracic glands, ecdysone, moult. Beside it is a box: 'High JH = stay larva/nymph; low JH = pupate and become adult'. I draw a butterfly cycle (egg, larva, pupa, adult) and a grasshopper cycle (egg, nymph, adult) with a photo of the exuvia from the school garden. My conclusion: An insect outgrows its hard shell by moulting in steps controlled by ecdysone, and juvenile hormone decides whether it only grows bigger or changes shape completely. A common misconception is that insects peel like skin or melt without control, but actually the exoskeleton is shed whole and every change is hormone-controlled.

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Claim	The claim fully answers both parts of the	The claim answers the driving question	The claim is vague, incomplete, or only

	driving question, names both growth patterns, and mentions the role of hormones in clear language.	but leaves out one idea, such as the hormones or one growth pattern.	partly related to the driving question.
Evidence	Uses specific evidence from the exuvia, the staircase and S-shaped graphs, and the life cycles, with accurate details such as the growth phases and the pupa stage.	Uses relevant evidence from most sources but some details are general or missing.	Evidence is limited, inaccurate, or not linked to the phenomenon.
Reasoning with Hormonal Control	Correctly explains the sequence of PTTH, prothoracic glands, ecdysone and the role of high versus low juvenile hormone, and links it clearly to the claim and evidence.	Explains most of the hormone sequence with a small error or a weak link to the evidence.	Hormone roles are missing, confused, or listed without explanation.
Annotated Graphical Model	Model is accurate, fully labelled, and shows both growth curves, the four sigmoid phases, the hormone sequence, and both types of metamorphosis.	Model includes most required parts with a few missing labels or small errors.	Model is incomplete, unclear, or contains major errors.
Scientific Communication and Context	Writing is clear and well organised, uses scientific terms correctly (ecdysis, exuvia, metamorphosis), uses Kenyan contexts, and corrects a misconception.	Writing is mostly clear and uses most terms correctly, with limited local context or no misconception addressed.	Writing is hard to follow, and scientific terms are missing or misused.

ATTRIBUTION AND LICENSING

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